Diseases of the Renal System

NURA 340 Pathophysiology | Lecture Study Guide

Instructor: Melissa Carson, SNM, MSN, RN

Lecture roadmap: Renal dysfunction overview → Acute kidney injury (prerenal, intrinsic, postrenal) → Chronic kidney disease → Glomerulonephritis → Nephrotic syndrome (and how it differs) → Nephrolithiasis (kidney stones)

This lecture applies normal renal physiology to kidney disorders. For each disorder, connect the cause and underlying mechanism to the symptoms and diagnostic findings, and explain why kidney problems can produce effects throughout the body. The instructor's recurring question is: where does the problem begin?

Where does the problem begin? Is it blood flow to the kidney (perfusion)? Injury within the kidney? Or obstruction of urine leaving the kidney? Then connect the problem to changes in filtration, fluid and electrolyte balance, and the findings you see.

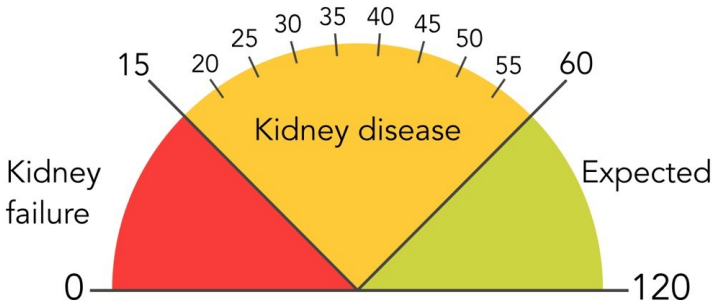
Remember: The kidneys do more than make urine. They regulate fluid, electrolytes, and acid-base balance, and they support blood pressure control, red blood cell production, and bone health. As you study each disorder, think about which function is disrupted and what you would expect to see because of it. Focus on what is covered in this lecture and on your Prepare page as you work through the ATI Engage modules.

Learning Objectives

- Etiology, pathogenesis and pathophysiology, manifestations, and diagnostic criteria of acute kidney injury.
- The same for chronic kidney disease.
- The same for glomerulonephritis.
- The same for nephrolithiasis.
- Correlate the pathophysiology of nephrotic syndrome with proteinuria, hypoalbuminemia, edema, and other clinical findings.

Introduction to Renal Dysfunction

Renal dysfunction means the kidneys are having difficulty carrying out one or more of their normal functions: waste removal, fluid and electrolyte regulation, acid-base balance, and hormonal functions.



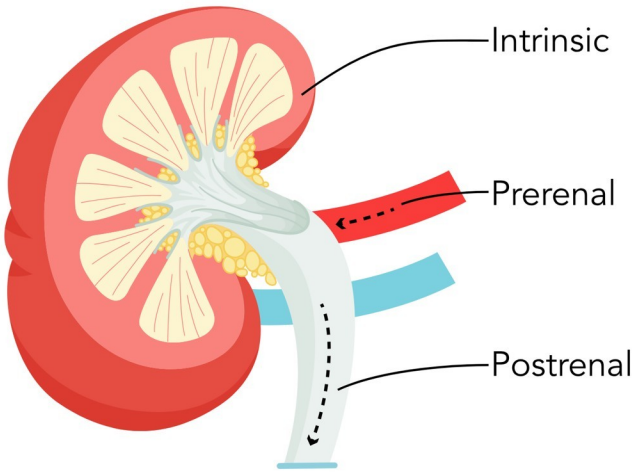
GFR scale: below 15 kidney failure, 15 to 60 kidney disease, above 60 expected (lecture slide)

Term	Meaning
Acute kidney injury (AKI)	Develops over a short period
Chronic kidney disease (CKD)	Abnormalities that persist for at least 3 months
AKI on CKD	A person can develop an acute kidney injury on top of chronic disease

GFR does not tell the whole story: Someone can have kidney damage, such as persistent albumin loss in the urine, even when GFR is relatively preserved. Keep two questions in mind: how well are the kidneys filtering, and is there evidence of kidney damage?

Acute Kidney Injury (AKI)

An abrupt decline in kidney function. Causes are organized by where the problem begins.



Prerenal, intrinsic, and postrenal causes of AKI (lecture slide)

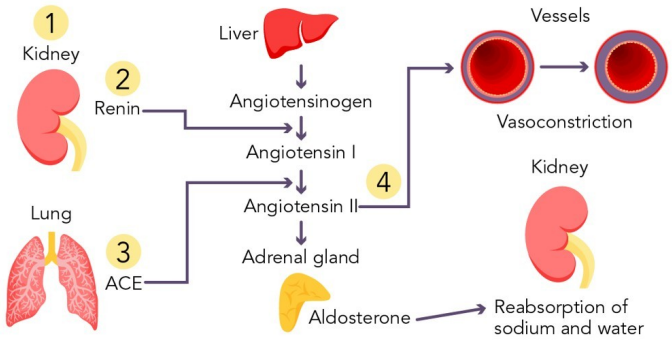
Etiology

Type	Where it begins	Examples and mechanism
Prerenal (before the kidney)	Inadequate blood flow reaching the kidneys; decreased perfusion without direct damage to the parenchyma	Significant or traumatic blood loss, dehydration. Initially the problem is perfusion, but prolonged poor perfusion deprives tissues of oxygen-rich blood and injures them
Intrinsic (within the kidney)	Direct damage to the tubules, glomeruli, blood vessels in the kidney, or surrounding tissues	Toxins, inflammation, sepsis, ischemia; diseases such as acute tubular necrosis (ATN), glomerulonephritis, and acute interstitial nephritis. In ATN, tubular cells are injured and cannot filter because of necrosis and tissue loss. Tubular injury interferes with reabsorption and secretion
Postrenal (after the kidney)	Urine flow is obstructed	Kidney stones, tumors, retroperitoneal fibrosis, enlarged prostate. Pressure builds upstream from the blockage and opposes filtration, and the kidneys become saturated with urine and fluid, causing structural injury

- **Risk factors:** age, pre-existing CKD, sepsis (a huge one), and exposure to nephrotoxic agents such as NSAIDs (for example ibuprofen), aminoglycosides, and iodinated contrast media. This is the connection to pharmacology.

Pathogenesis and Pathophysiology

- **Prerenal:** reduced perfusion activates compensatory mechanisms such as RAAS, which conserves sodium and water to maintain blood pressure.



RAAS: renin, angiotensin I and II, ACE, aldosterone (lecture slide)

- **Intrinsic:** direct injury from ischemia, toxins, or inflammation damages parts of the kidney and lowers GFR, impairing kidney function.
- **Postrenal:** obstruction of urine flow raises pressure inside the kidney, slows filtration, and then causes structural injury.

Manifestations

Manifestation	Why
Oliguria or anuria	Not enough urine (oliguria) or no urine (anuria) produced
Peripheral edema	The body cannot get rid of the water, so it retains it
Rapid weight gain	Fluid retention. This is not 10 pounds in a month; it is a very rapid gain over a day or two (or up to a week) because the body cannot "pee off" the water
Encephalopathy, confusion, lethargy	Waste products build up and cause systemic effects (uremia)

Know the baseline: It is important to understand the person's baseline kidney function. Also, AKI can occur even when urine output stays relatively normal, so assess the overall picture, not one finding.

Diagnostic Criteria

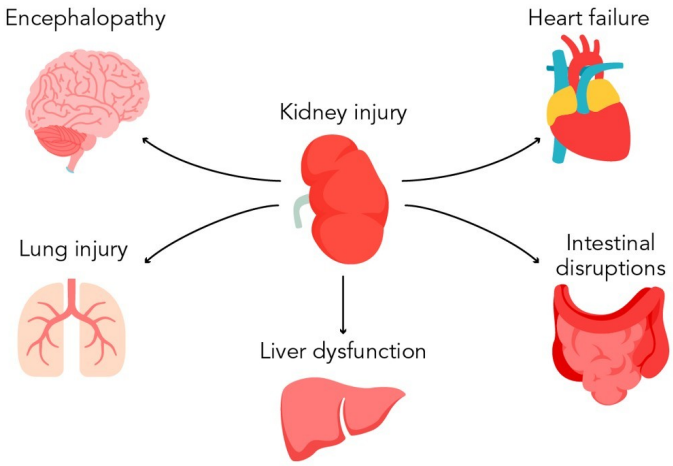
Important to understand, but you will not be making this diagnosis. Either creatinine or urine output can identify AKI; you do not have to wait for both.

Criterion (any one)	Detail
Creatinine rise	At least 0.3 mg/dL within 48 hours (needs a baseline)
Creatinine rise	At least 1.5 times baseline within 7 days
Urine output	Below 0.5 mL/kg/hour for at least 6 hours (used when there are no baseline labs)

- **Labs and tests:** serum creatinine, BUN, electrolytes (potassium or other substances accumulating), bicarbonate and ABG (acid-base effects), urinalysis (clues about tissue injury), and renal ultrasound to identify obstruction, especially if the cause is unclear. If AKI is suspected, get labs as soon as possible, then a series of labs.

Impact on the Health of the Client

- **Main physiological concerns:** fluid overload, hyperkalemia, metabolic acidosis, and waste accumulation.

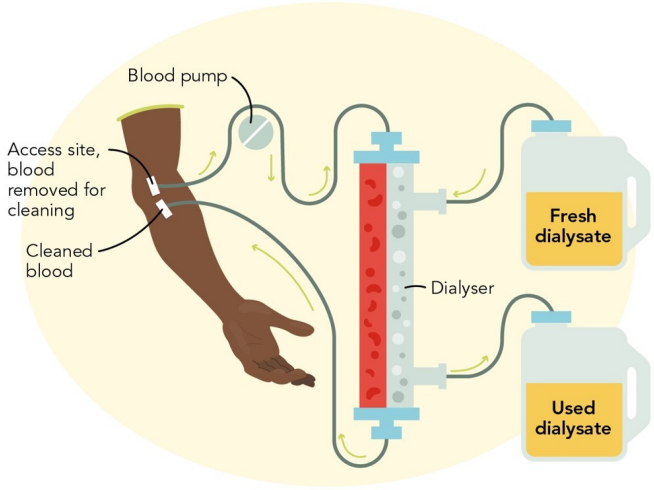


Systemic effects of kidney injury: brain, heart, lungs, liver, intestines (lecture slide)

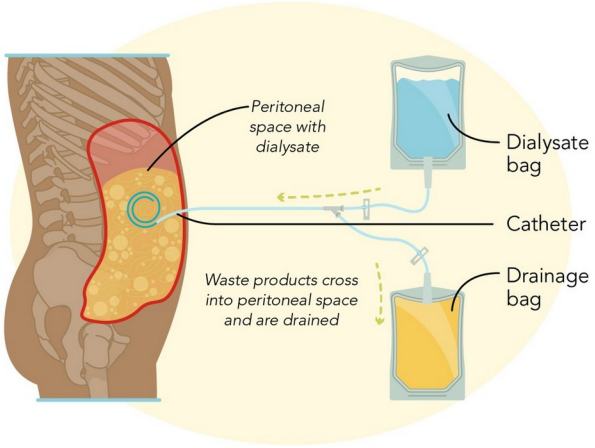
- **Systems affected:** severe dysfunction can affect the lungs, heart, brain, gastrointestinal system, and liver, so it is taken very seriously.
- **Short term:** acid-base disturbance (the kidneys cannot help regulate it), electrolyte imbalances, fluid overload, inflammation, and uremia.
- **Long term:** if untreated or improperly managed, permanent loss of nephrons, reduced filtration capacity, and long-term decline in kidney function.
- **Recovery:** some people recover kidney function and others have lasting damage. It depends on the extent of injury, how long function is impaired, the cause (an underlying disease brings its own consequences), age, and overall health.

Management (awareness only)

Nursing interventions are not covered. The goal for prerenal AKI is to restore blood flow. For postrenal AKI, it is to clear the blockage and get urine flowing again. Dialysis is needed in some severe cases and is not an expected outcome of all AKI. ATI covers the three main types: hemodialysis, peritoneal dialysis, and continuous renal replacement therapy (CRRT). These come later.



Hemodialysis circuit (lecture slide)



Peritoneal dialysis (lecture slide)

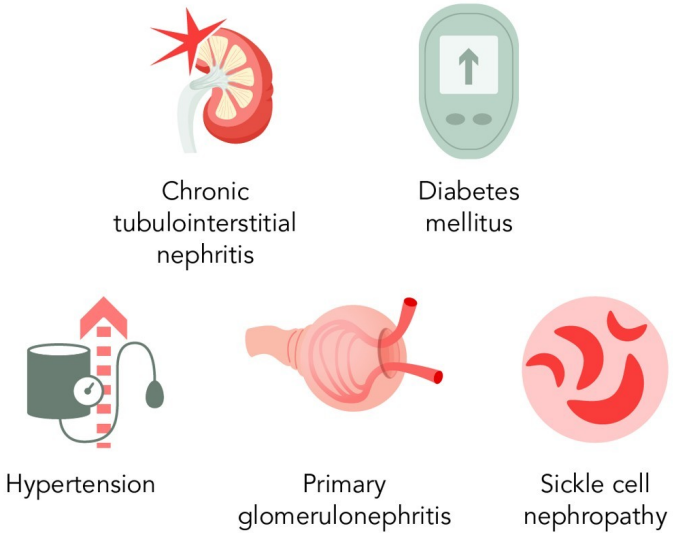
Discussion Question 1: Explain the etiology of prerenal, intrinsic, and postrenal AKI

Try explaining it to a patient or a friend or family member outside healthcare. If you can explain it to someone else, you understand it. Use the Etiology table above: prerenal is a blood flow (perfusion) problem, intrinsic is tissue injury inside the kidney, and postrenal is an obstruction after the kidney.

Chronic Kidney Disease (CKD)

Persistent kidney damage or reduced function lasting more than 3 months. This is a really important distinction from AKI.

Etiology



Common causes of CKD (lecture slide)

- **Common causes:** diabetes mellitus and hypertension (most common), plus glomerular disease (primary glomerulonephritis), chronic tubulointerstitial nephritis, sickle cell nephropathy, and other conditions.

Pathogenesis and Pathophysiology

1. **Nephron loss:** as functioning nephrons are lost, the remaining nephrons take on more work to compensate.
2. **Early benefit:** this compensation can help preserve filtration at first.
3. **Later harm:** increased pressure and workload contribute to further injury over time. The remaining nephrons are not designed for the increased workload, and their function declines.

Connect each finding to a normal kidney function that has been impaired:

Impaired function	Result
Sodium and water excretion	Retention of sodium (water follows). Too much fluid in the vasculature causes edema and hypertension
Acid excretion	Metabolic acidosis
EPO production	Less EPO tells the bone marrow to make red blood cells, causing anemia
Vitamin D activation	Mineral and bone loss
Urea excretion	Urea is excreted through sweat instead, changing body odor and leaving uremic frost as sweat dries

Manifestations

- **Early:** few or no noticeable symptoms. As function declines, effects become more obvious and broader.
- **Progressive:** fatigue, anorexia, nausea, vomiting, sleep disturbances, cognitive impairment, muscle cramps, pruritus (itching), and cardiopulmonary manifestations.
- **Uremic frost:** a white residue on the skin as urea-containing sweat dries. A sign of really late-stage kidney disease.

Diagnostic Criteria

Measurement	What it shows
eGFR	Estimates filtration
Urine albumin-to-creatinine ratio (ACR)	Identifies damage to the filtration barrier. Everyone with suspected CKD should have an initial ACR on a urine specimen

Exam note: The two measurements give different information, so consider them together. For CKD, the abnormality must persist for at least 3 months. One abnormal result during an acute illness does not by itself establish chronic disease. CKD is diagnosed if GFR is below 60 for at least 3 months, or if there are markers of kidney damage such as proteinuria or hematuria. Someone can have CKD with a relatively preserved GFR if persistent markers of damage are present.

- **Staging:** ranges from G1 to G5 (G5 is end-stage renal disease). You are not expected to know the staging; only know the sequence.

- **Imaging and biopsy:** ordered when more detail is needed about the cause, extent, or type of damage.

Impact on the Health of the Client

- **Complications:** fluid overload, electrolyte disturbances, and the other manifestations above as function declines. Anemia and cardiovascular complications (high blood pressure, dysrhythmias, atherosclerosis) are more likely, and overall cardiovascular risk is greatly increased.
- **Progression:** varies. Some people stay stable for a long time; others develop advanced kidney failure quickly.
- **Management (awareness only):** slow progression, treat symptoms as they arise, and remove substances the body cannot excrete. Nursing interventions are not covered.

Discussion Question 2: Why is untreated CKD a risk for cardiovascular disease?

Be able to explain this using what the kidneys do: sodium and water retention raise blood volume and blood pressure (the vicious cycle of kidney disease and hypertension from the first lecture), and CKD also brings anemia, acidosis, and electrolyte changes that affect the heart. These show up as hypertension, dysrhythmias, and atherosclerosis.

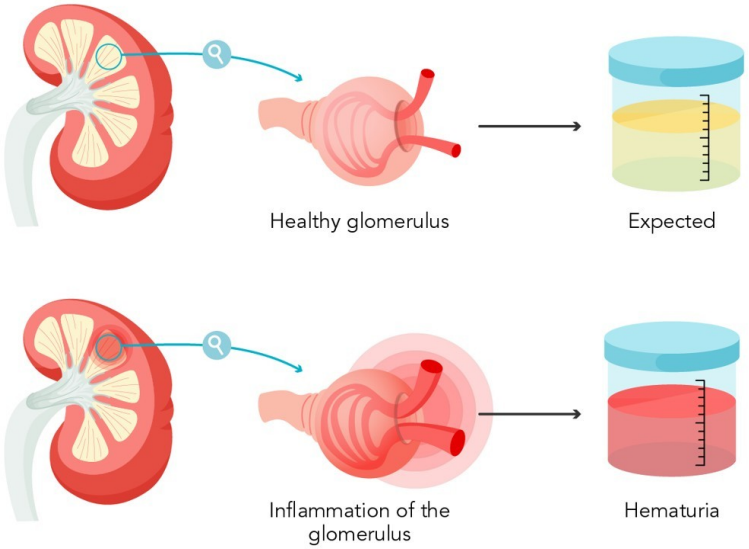
Inflammatory and Obstructive Renal Disorders

Two specific mechanisms: glomerular injury and urinary obstruction. As you move through these, ask: is this a filter problem or a drainage problem?

Disorder	What it is
Glomerulonephritis	Inflammation of the kidney's filters. It affects how well they filter and what they allow into the urine. If only certain things reach the urine, ask what is being left in the blood
Nephrotic syndrome	A pattern dominated by excessive protein loss through the filtration barrier. The barrier should keep blood cells and protein in the blood
Renal calculi (nephrolithiasis)	Kidney stones formed by substances in the urine. They may remain in the kidney or move into the urinary tract, where they can obstruct flow and contribute to kidney injury

Glomerulonephritis

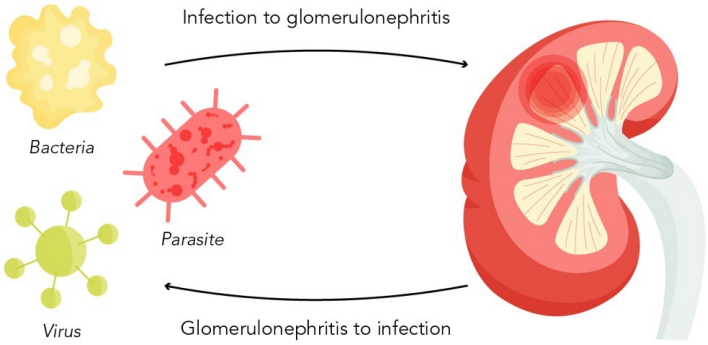
The glomerular filtration barrier normally keeps blood cells and most protein in the bloodstream. When the barrier is damaged, blood cells or large amounts of protein may appear in the urine, and the pattern of what leaks through helps identify the disorder.



Healthy glomerulus (expected urine) vs. inflamed glomerulus (hematuria) (lecture slide)

Etiology

- **Immune activity:** often involves immune complex deposition or antibodies that target structures of the filtration barrier.
- **After infection:** can follow a skin infection, viral infection, or other illness. A preceding infection does not mean bacteria are directly infecting the glomeruli; the damage is from the inflammation process secondary to infection.
- **Autoimmune and other causes:** lupus, minimal change disease in children, and (less commonly) diabetes and amyloidosis.



Infection can lead to glomerulonephritis, and glomerulonephritis to infection (lecture slide)

Pathogenesis and Pathophysiology

1. **Inflammation:** damages the filtration barrier (tears and breakdown of the filter) and reduces filtration.
2. **Leakage:** red blood cells enter the urine (hematuria), and protein may also leak through.
3. **Retention:** reduced filtration promotes sodium and water retention, explaining edema and hypertension.

This combination (hematuria, impaired filtration, fluid retention) is the nephritic pattern. Severity varies, and some glomerular diseases overlap nephritic and nephrotic findings.

Manifestations

- **Systemic:** fatigue, weakness, loss of appetite, nausea, and vomiting.
- **Urine:** blood and protein in the urine; urine may look brown, tea-colored, or cola-colored.
- **Fluid:** edema (face and periorbital swelling, especially early in the morning) and high blood pressure because inflamed kidneys hold on to salt and water.
- **Waste:** the kidneys do not clean the blood as well, so waste products build up.

Diagnostic Criteria

Test	Finding
Urinalysis	Blood in the urine. Red blood cell casts are particularly useful because they suggest bleeding originating within the kidney, often from glomerular inflammation
Proteinuria	The filtration barrier is letting protein escape
Creatinine	Rising creatinine suggests reduced filtration
Other blood tests	Complement levels, evidence of a preceding infection, autoantibody tests (autoimmune cause), and hepatitis tests (infection-related kidney disease)
Biopsy	May identify specific diseases; later in the workup or in severe or unclear cases

Impact on the Health of the Client

- **Complications:** electrolyte imbalances, particularly hyperkalemia and hyponatremia, and cardiovascular disease.
- **Progression:** glomerulonephritis can cause AKI. If the injury persists or leads to scarring and loss of function, it can contribute to CKD.

Management is not covered.

Discussion Question 3: Why does glomerulonephritis often present with hypertension or edema?

- **Inflammation:** reduces filtration and promotes sodium and water retention. More fluid means edema, and more volume in the vessels can lead to hypertension.
- **Protein loss:** proteins spill into the urine because the barrier has lost its integrity. When protein loss is substantial, plasma oncotic pressure falls, which further contributes to edema.

Nephrotic Syndrome

Pathway on the slide: Damage to the glomerular filtration barrier → heavy protein loss in urine → low blood albumin → fluid shifts into tissues → edema.

A group of findings that happen when the kidney's filter is damaged. The damaged barrier allows albumin to leak into the urine. The instructor's image: a strainer with holes that got too big.

Finding	Mechanism
Heavy proteinuria	Increased permeability of the glomerular filtration barrier. Urine may look foamy (a clue to remember)
Hypoalbuminemia	So much protein (particularly albumin) is lost that blood levels fall
Edema	Albumin normally helps keep fluid inside the blood vessels. When it drops, fluid leaks into the tissues. The kidneys also hold on to extra salt and water, which worsens the swelling. Seen in the legs, around the eyes (periorbital), and in the belly
High lipids	The body responds to protein loss by making more fats, so blood lipids and cholesterol rise. This is a compensatory change

Four findings from one cause: Heavy protein in the urine, low blood albumin, swelling, and high lipids all come from a damaged kidney filter.

Nephrotic Syndrome vs. Glomerulonephritis

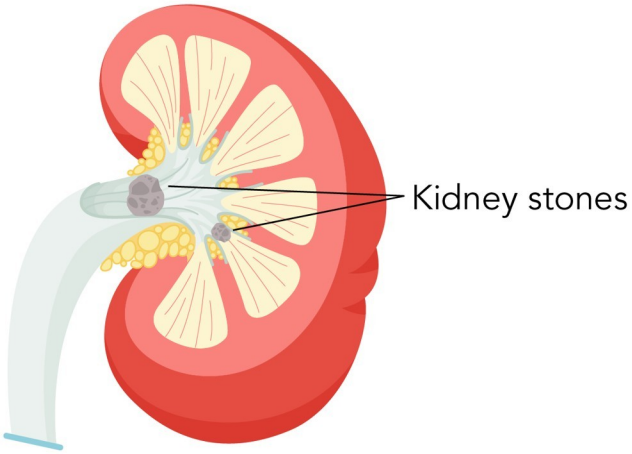
Both involve a damaged filter. The easiest way to tell them apart: nephrotic is a leaky filter; glomerulonephritis (nephritic) is an inflamed filter.

Lab / Finding	Nephrotic Syndrome (leaky filter)	Glomerulonephritis (inflamed filter)
Urine protein	Heavy (3.5 g/day or more)	Mild to moderate
Urine blood	Minimal or absent	Hallmark (RBC casts, dysmorphic RBCs)
Blood albumin	Low (hypoalbuminemia)	Usually normal
Blood lipids	Elevated	Normal
Blood pressure	Often normal early	Frequently high
Kidney function (GFR/creatinine)	Usually preserved early	Often declining
Edema	Yes (legs, periorbital, abdomen)	Yes (face, periorbital, especially in the morning)
Urine appearance	Foamy	Brown, tea-colored, or cola-colored

Edema appears in both. Make sure you know the difference.

Nephrolithiasis (Kidney Stones)

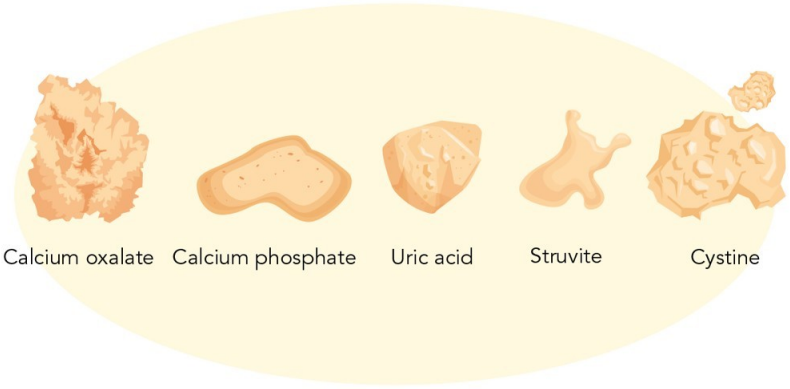
ATI uses the term nephrolithiasis for renal calculi. Stones are hardened mineral deposits that form within the kidneys.



Kidney stones in the kidney (lecture slide)

Etiology

- **Imbalance:** stones develop when the usual balance of water, salt, and minerals in the urine is out of balance and the urine becomes more concentrated.
- **Risk factors:** being male, not drinking enough water, too much salt and animal protein, diabetes or metabolic syndrome, family history, and certain medications.



Stone types: calcium oxalate, calcium phosphate, uric acid, struvite, cystine (lecture slide)

Exam note: Calcium-containing stones (calcium oxalate and calcium phosphate) are the most common. Other types are uric acid, struvite, and cystine. Urine chemistry determines which stone type forms.

Pathogenesis and Pathophysiology

1. **Crystallization:** particles in concentrated urine crystallize. Low urine volume (for example from dehydration) increases concentration.
2. **Growth:** the crystals aggregate and grow into a stone.
3. **Movement:** a stone may stay in the kidney without symptoms. If it moves into a ureter and obstructs flow, pressure builds upstream. The ureter stretches and contracts, causing severe pain.

4. **Backup:** if obstruction persists, hydronephrosis can develop (urine backs up because the kidney cannot drain into the bladder, damaging the kidney). It can also stretch the renal capsule and cause severe pain.

Manifestations

Manifestation	Why
Renal colic (flank pain)	Sudden, severe, sharp pain from ureter stretch and contractions. Radiates to the abdomen, groin, or inner thigh as the stone moves
Hematuria	Irritation by the stone
Urgency and frequency	Stone nearer the bladder irritates it
Nausea, vomiting, sweating	Visceral pain

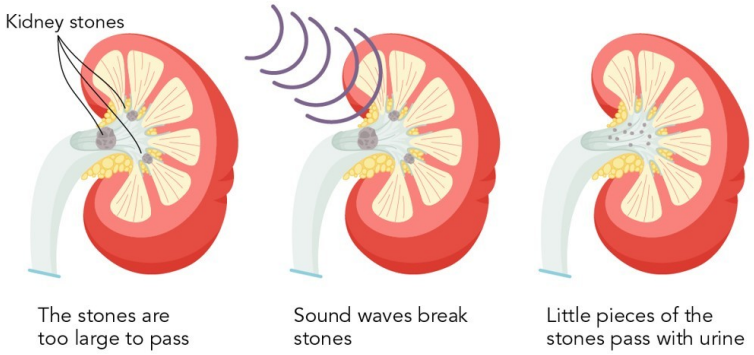
Not every stone causes obstruction. It depends on the size of the stone, and not every stone produces the same symptoms.

Diagnostic Criteria

- **Urinalysis:** blood and crystals in the urine.
- **Blood:** creatinine and urea to assess kidney function. Frequent stone formers may have additional testing that is not covered.
- **Imaging:** a CT or ultrasound, depending on the patient, if someone presents to the emergency room with symptoms.

Impact and Management (awareness only)

Focus on crystal formation, obstruction, pain, and the fact that stones can cause kidney injury. Smaller stones may pass on their own with urination, but size alone does not guarantee passage. The slide shows that large stones may be broken with sound waves (lithotripsy) so small pieces pass with urine. Management is not covered.



Stones too large to pass, sound waves break stones, pieces pass with urine (lecture slide)

Discussion Question 4: Why do kidney stones raise the risk of high blood pressure or heart disease?

The instructor said this one takes more thought: consider which mechanisms that lead to a kidney stone might also relate to high blood pressure and heart disease, or the reverse. Think about the earlier links between kidney function, sodium and water balance, and blood pressure, and between obstruction, kidney injury, and CKD.

Wrapping Up

As you review, start with the mechanism:

- **AKI:** does the problem begin with perfusion, tissue injury, or obstruction?
- **CKD:** connect the persistent damage to the loss of normal kidney function.
- **Glomerular disorders:** what is leaking into the urine, and how is filtration impaired?
- **Calculi:** follow the effects of crystal formation and obstruction.

Quick Review

Condition	Key points
AKI	Abrupt decline; prerenal (perfusion), intrinsic (ATN, glomerulonephritis, interstitial nephritis, toxins, sepsis), postrenal (obstruction: stones, tumor, BPH). Oliguria/anuria, edema, rapid weight gain, encephalopathy. Creatinine up 0.3 in 48 h, or 1.5x in 7 days, or urine output under 0.5 mL/kg/h for 6 h. Risks: fluid overload, hyperkalemia, metabolic acidosis
CKD	More than 3 months; diabetes and hypertension most common. Remaining nephrons overwork. Edema, hypertension, acidosis, anemia (low EPO), bone loss (low vitamin D), uremic frost late. eGFR below 60 and/or ACR (proteinuria, hematuria) for 3 months. High cardiovascular risk
Glomerulonephritis	Inflamed filter; immune-mediated, often after infection or autoimmune (lupus). Hematuria (cola-colored, RBC casts), mild to moderate proteinuria, edema, hypertension, rising creatinine. Can cause AKI and CKD; hyperkalemia and hyponatremia
Nephrotic syndrome	Leaky filter; heavy proteinuria (3.5 g/day or more, foamy urine), low albumin, edema (periorbital, legs, abdomen), high lipids; blood pressure often normal early
Nephrolithiasis	Concentrated urine crystallizes; calcium oxalate and phosphate most common; flank pain (renal colic) radiating to groin, hematuria, nausea; obstruction can cause hydronephrosis and kidney injury; UA, CT or ultrasound

Notes on Transcript Cleanup

Corrected: "R**S" is written as RAAS; "postnatal" (a spoken slip) is written as postrenal; "uremic acid" is written as uremia and waste buildup; "electrical imbalances" is written as electrolyte imbalances. The transcript says AKI causes "metabolic acidosis or alkalosis"; the guide lists metabolic acidosis, which is the expected finding. Where the instructor only said to pause and think (Questions 1, 2, and 4), the guide gives a short summary built from material she covered; she did not give a worked answer for those. The lecture speaker is not named on the title slide; the byline follows the previous renal lecture.